



[illegible]

CPU Address Bus Buffering

CPU Address Bus Nybble Decoding

Note:
The high nybbles are not taking /ROMSEL or A15 into account, so they are 32kB mirrored at this point.
"Reg_Access" is included in downstream logic to specify which 32kB region.

M2 Spaghetti That Goes Nowhere

CPU_R/W Pin

ROMSEL Pin

Sheet: /CPU Address Bus/
File: cpuAddressBus.kicad_sch

Title:
Size: USLdger Date:
KiCad E.D.A. 9.0.9

Rev:
Id: 2/25

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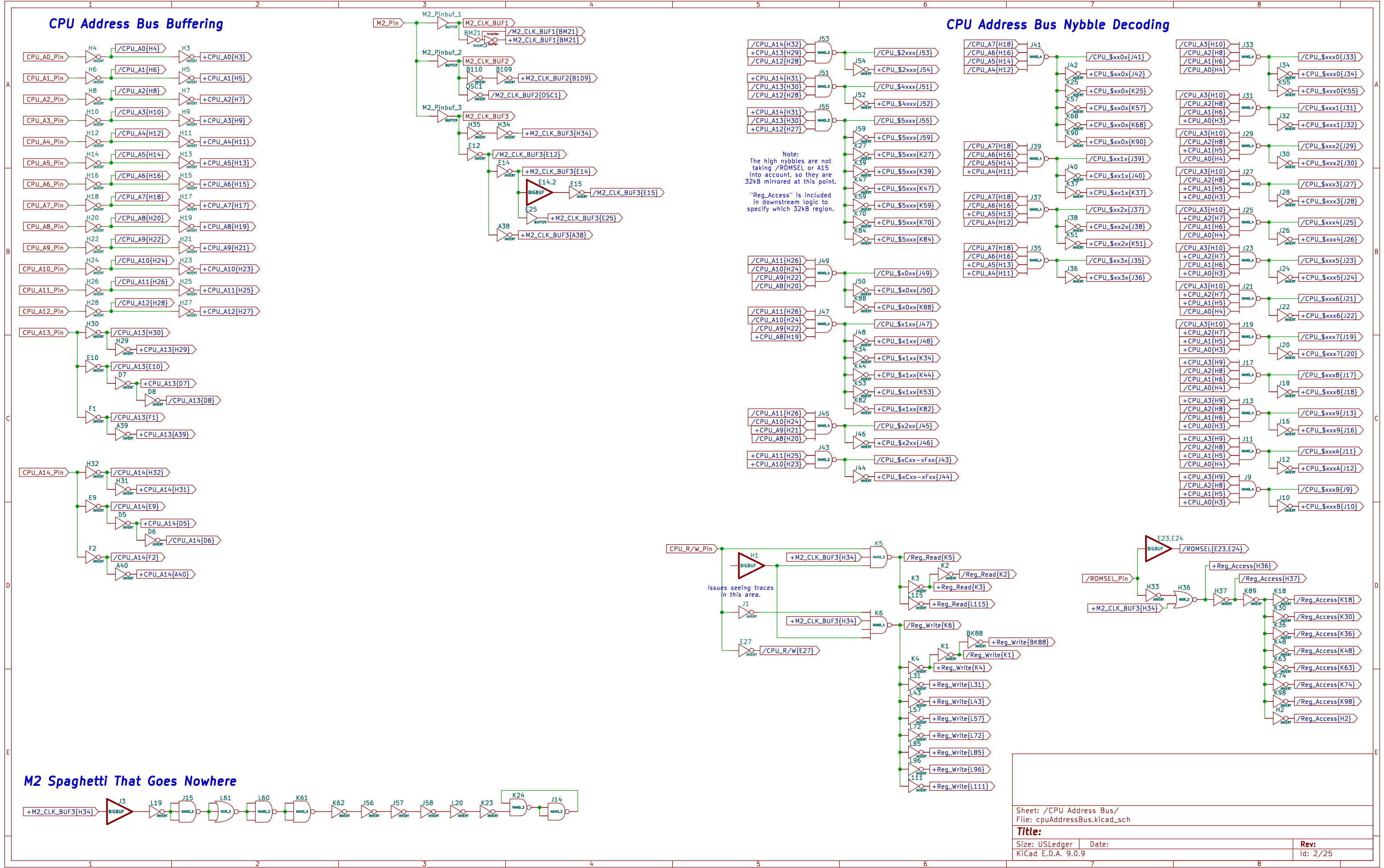
ROMSEL and Register Access

Register Access and CPU R/W Pin

Issues seeing traces in this area.

Sheet: /CPU Address Bus/ File: cpuAddressBus.kicad_sch

Title:	
Size: USLdger	Date:
KiCad E.D.A. 9.0.9	Rev:
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CPU_R/W Pin Logic

ROMSEL Pin Logic

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Title:

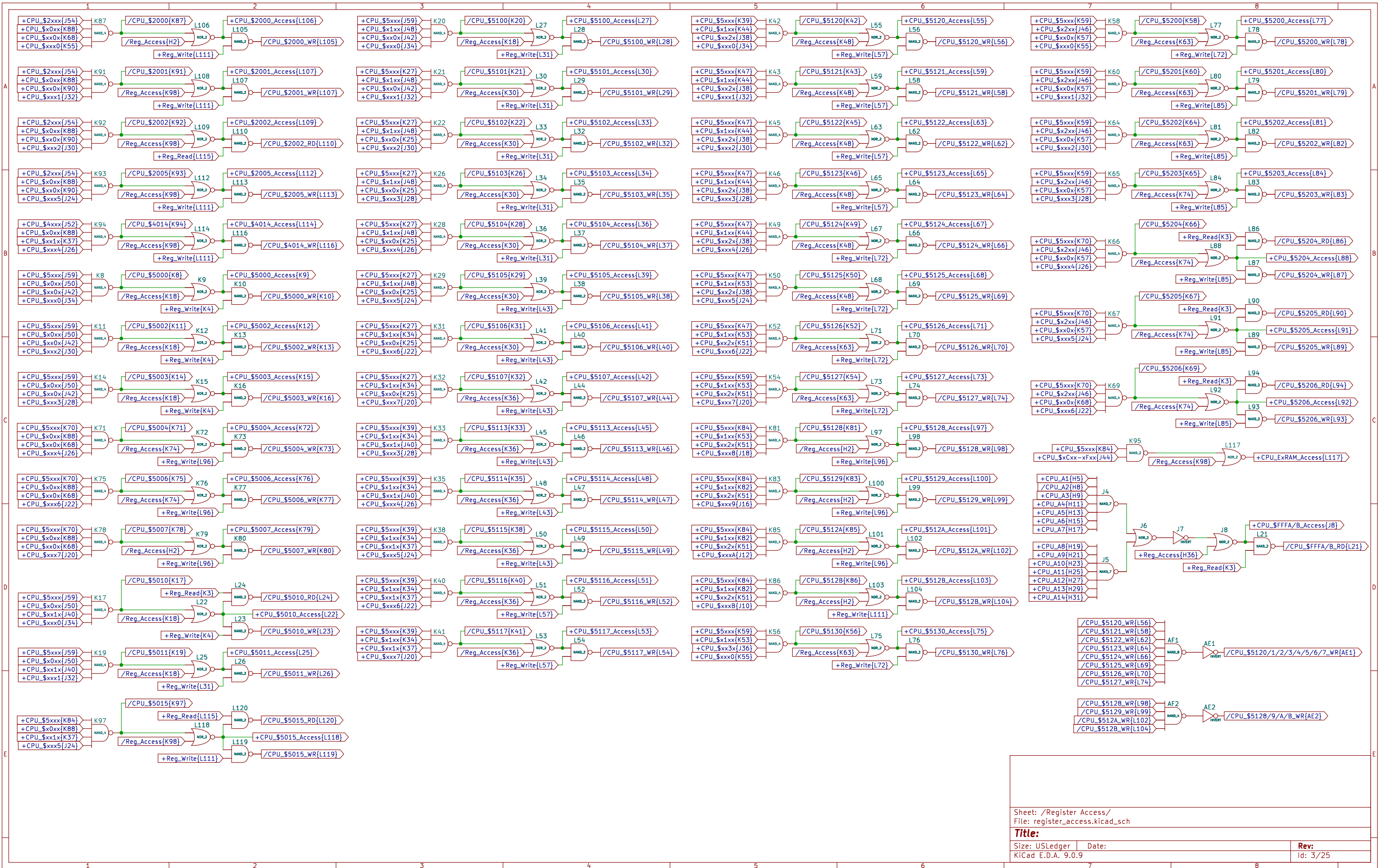
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Rev:

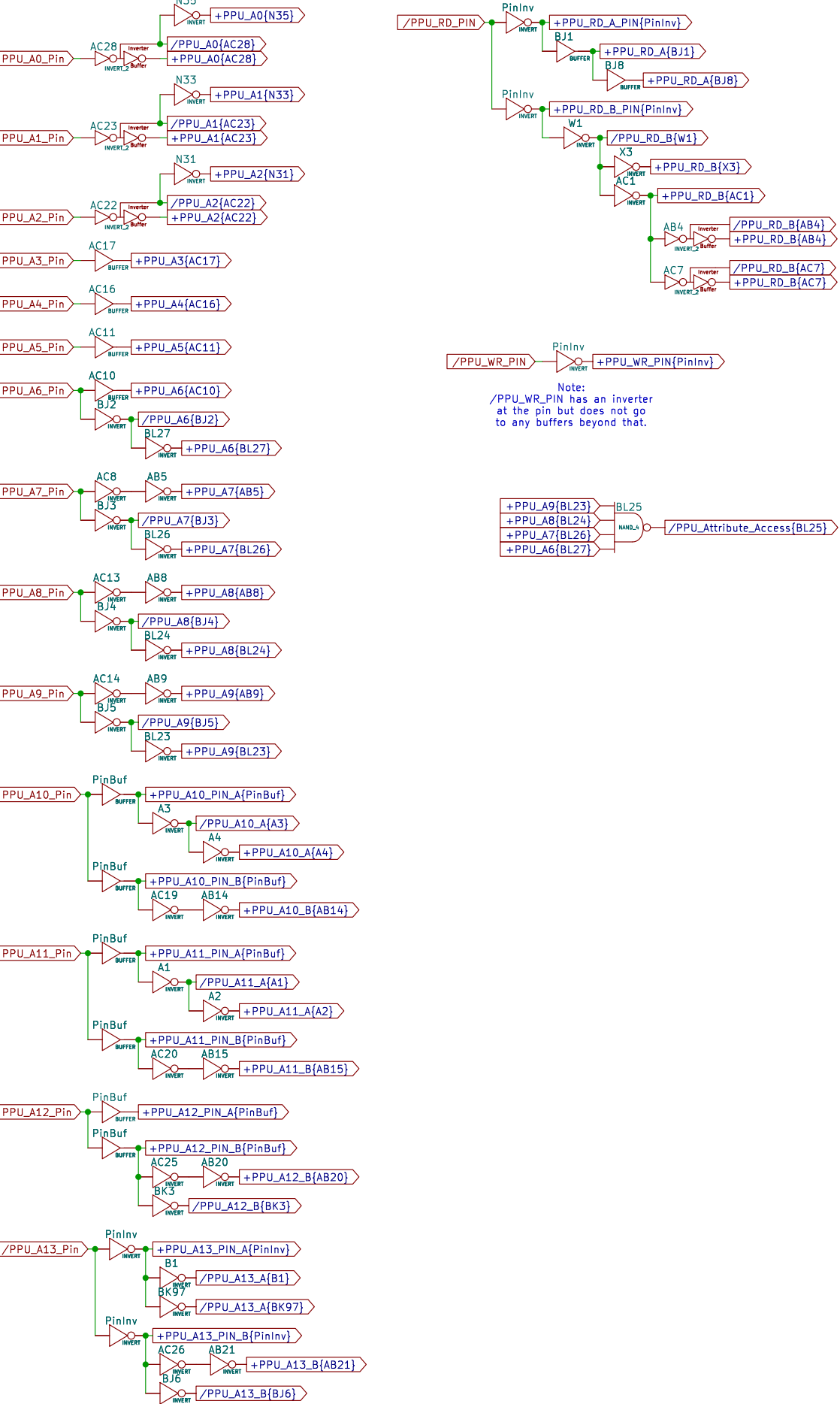
KiCad E.D.A. 9.0.9

Id: 2/25

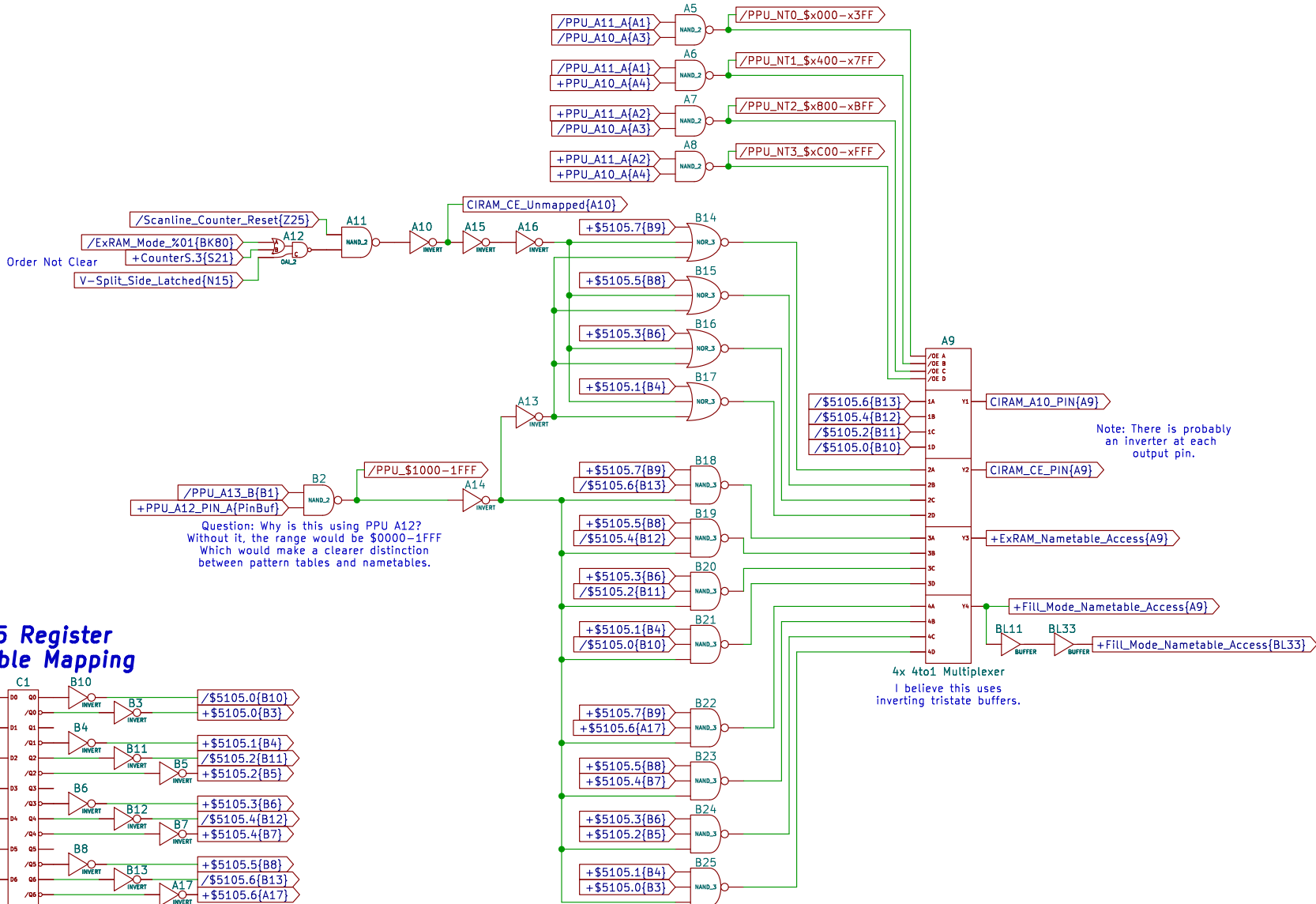
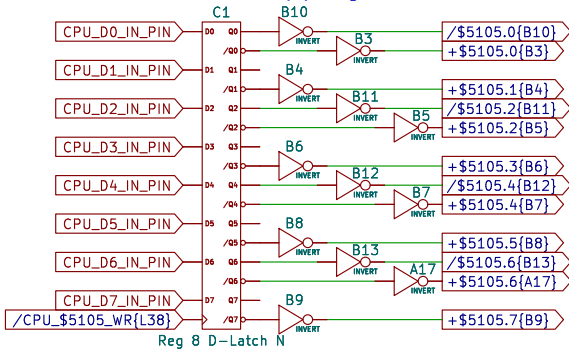
[illegible]



PPU Address Bus Buffering



\$5105 Register Nametable Mapping



Sheet: /Nametable Mapping/
File: nametable_mapping.kicad_sch

Title:

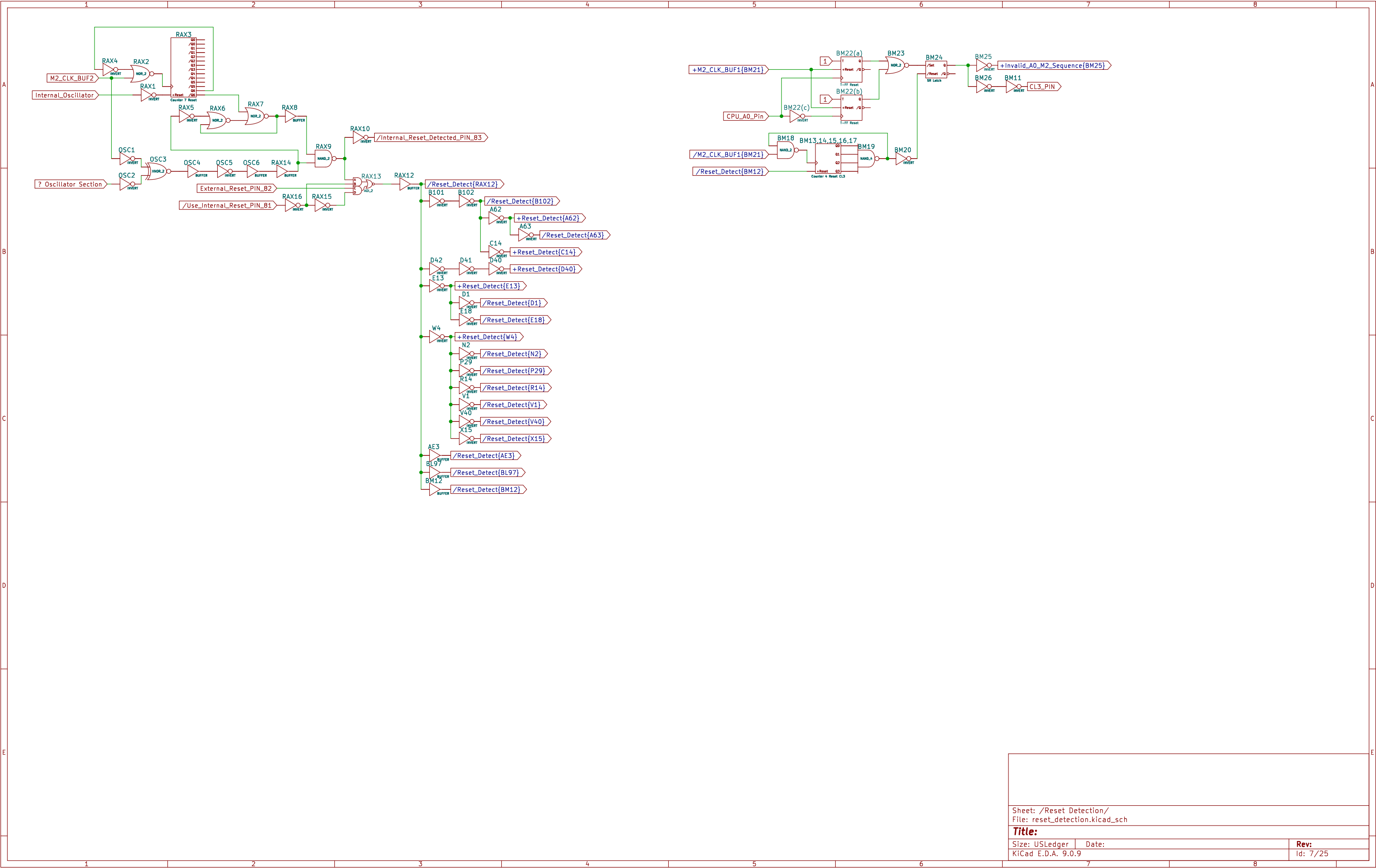
Size: USLedger

Date:

Rev:

KiCad E.D.A. 9.0.9

Id: 5/25



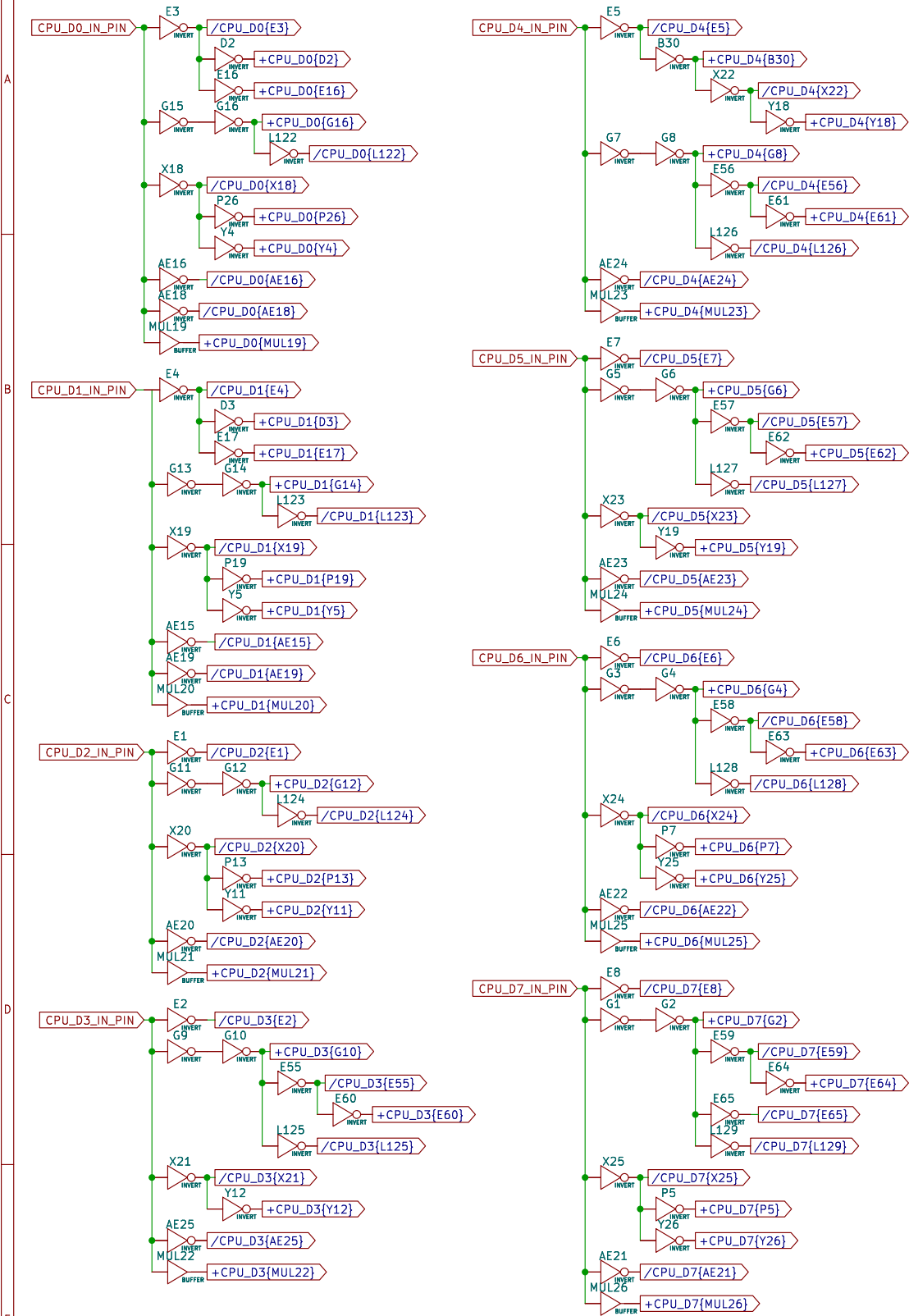
Sheet: /Reset Detection/
File: reset_detection.kicad_sch

Title:

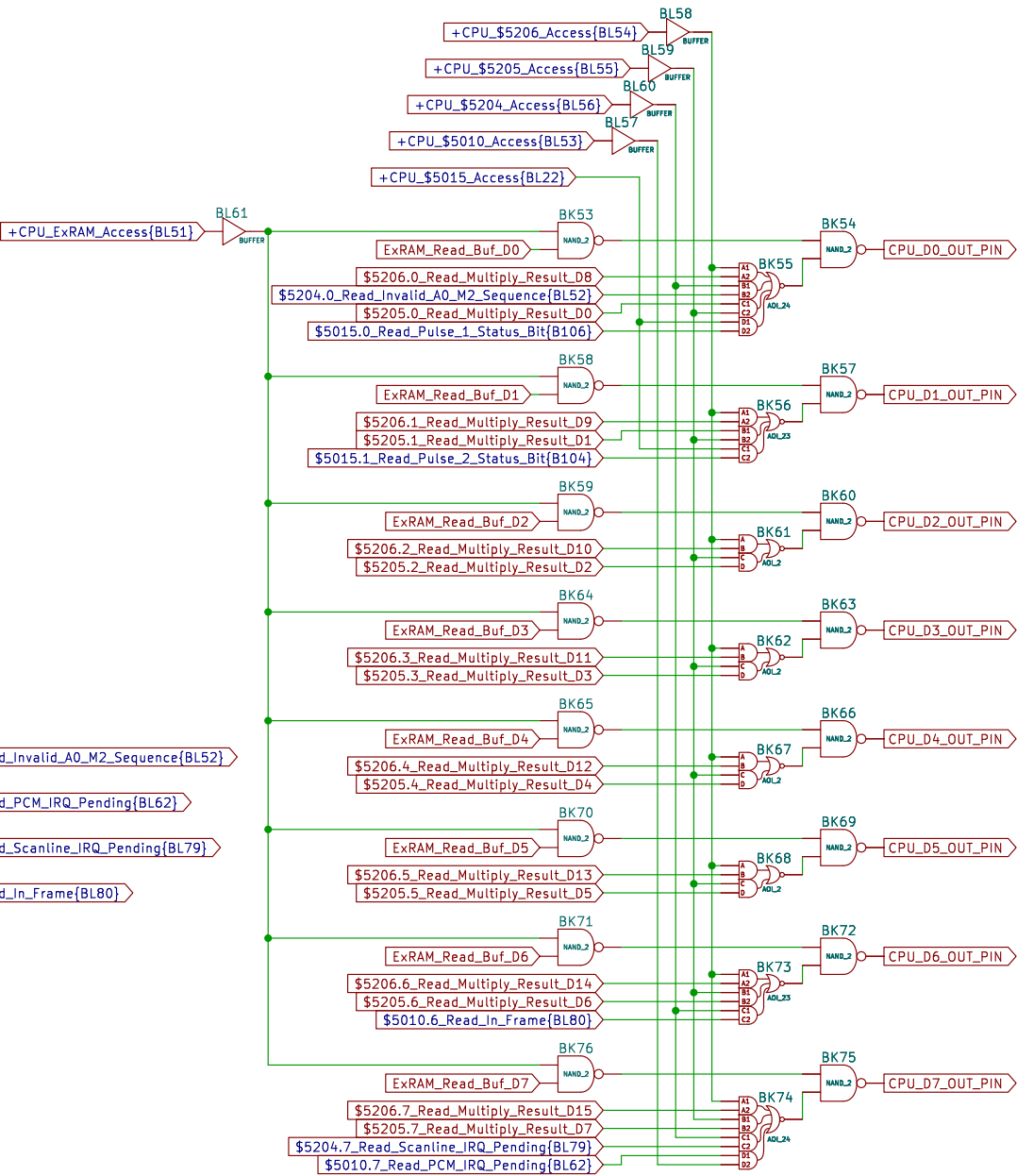
Size: USLedger | Date:
KiCad E.D.A. 9.0.9

Rev:
Id: 7/25

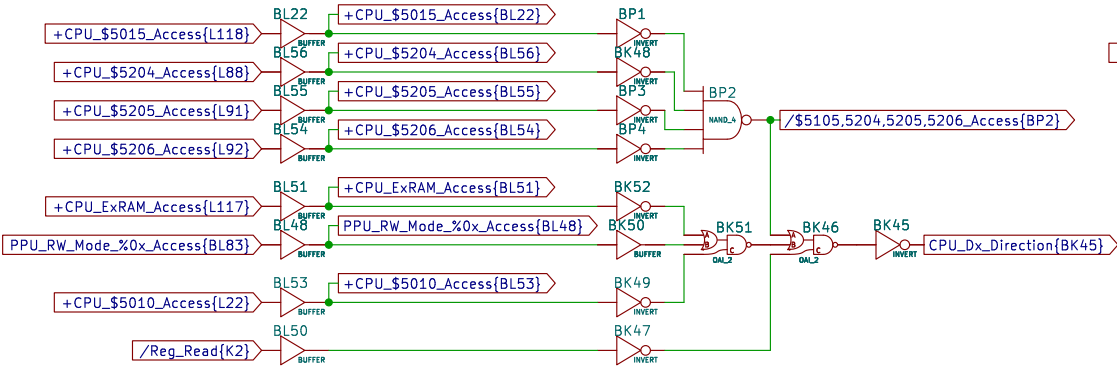
Incoming CPU Data Bus Buffering Bonanza



Multiplexing Data to be Driven onto the CPU Data Bus



Decides When to Drive the CPU Data Bus



\$5010 Register Mode & IRQ Enable

Read Mode:

- Inputs: /CPU_A14{H32}, /Reg_Access{H37}, +Reg_Read{L115}, +\$5010.0_{(PCM_READ_MODE)}{K101}, /CPU_\$5010_RD{L24}.
- Logic: /CPU_A14{H32} and /Reg_Access{H37} are NANDed (K96) and inverted (K99). The result is ANDed with +Reg_Read{L115} (K102) and +\$5010.0_{(PCM_READ_MODE)}{K101} (K103). The output is inverted (K108) and connected to the /Data_Bus=\$00{L130}.

Write Mode:

- Inputs: +CPU_D0{G16}, +CPU_D7{G2}, /Reset_Detect{B102}, /CPU_\$5010_WR{L23}, /Data_Bus=\$00{L130}, /Reset_Detect{B102}.
- Logic: +CPU_D0{G16} and +CPU_D7{G2} are NANDed (K104) and inverted (K105). The result is ANDed with /Reset_Detect{B102} (K106) and /CPU_\$5010_WR{L23} (K107). The output is inverted (K108) and connected to the /Data_Bus=\$00{L130}.

DAC Output:

- Inputs: /Data_Bus=\$00{L130}, /CPU_D0{G16}, /CPU_D1{G14}, /CPU_D2{G12}, /CPU_D3{G10}, /CPU_D4{G8}, /CPU_D5{G6}, /CPU_D6{G4}, /CPU_D7{G2}.
- Logic: The /Data_Bus=\$00{L130} is connected to the D0-D7 inputs of the DAC. The DAC outputs are PCM_DAC_Bit_0 through PCM_DAC_Bit_7.

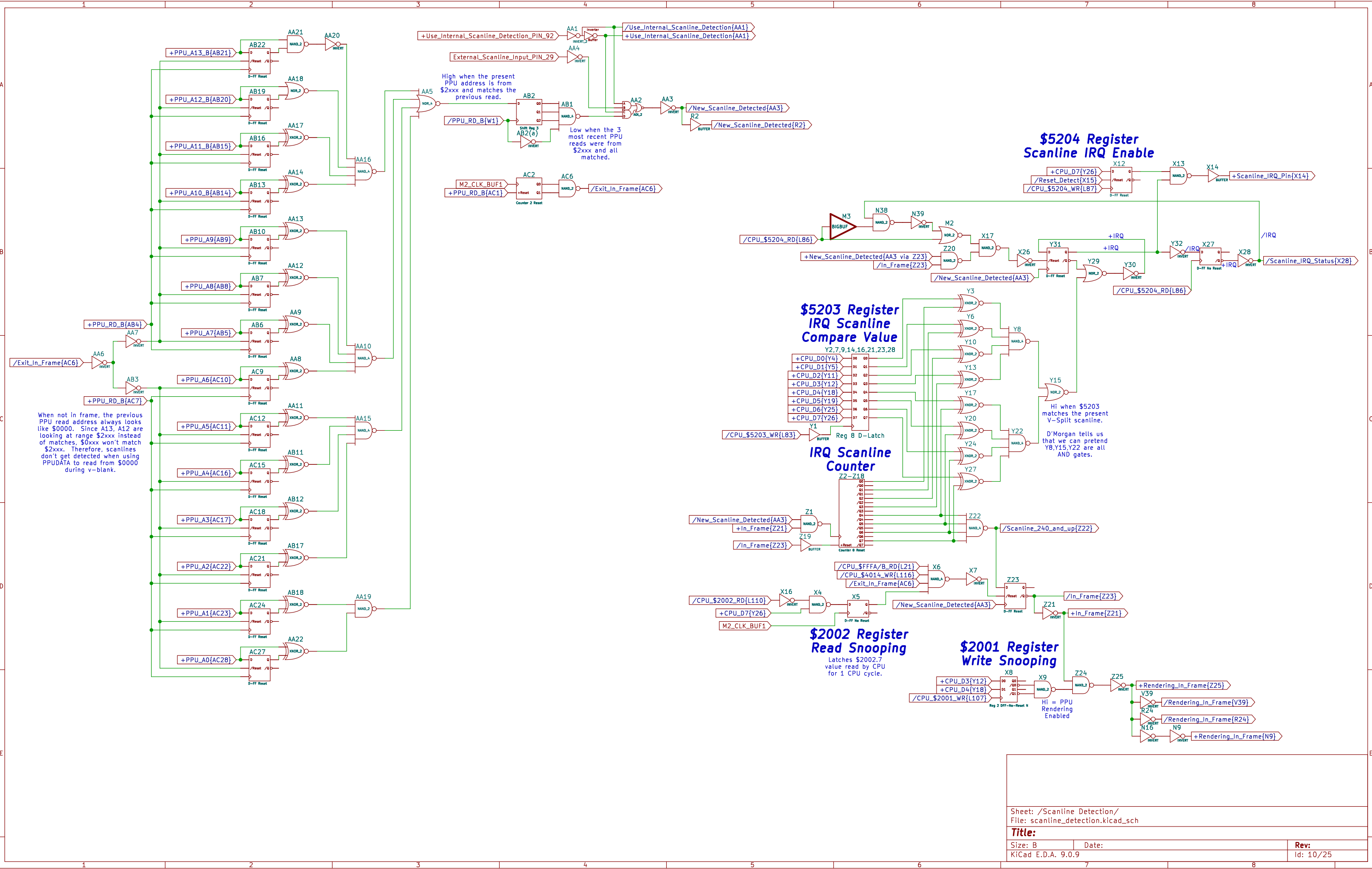
IRQ Enable:

- Inputs: /Data_Bus=\$00{L130}, /\$5010.7_{(PCM_IRQ_ENABLE)}{K101}, /Reset_Detect{B102}.
- Logic: /Data_Bus=\$00{L130} and /\$5010.7_{(PCM_IRQ_ENABLE)}{K101} are NANDed (K112) and inverted (K113). The result is ANDed with /Reset_Detect{B102} (K114) and inverted (K114) to produce the +PCM_IRQ_If_Enabled{K114} output.

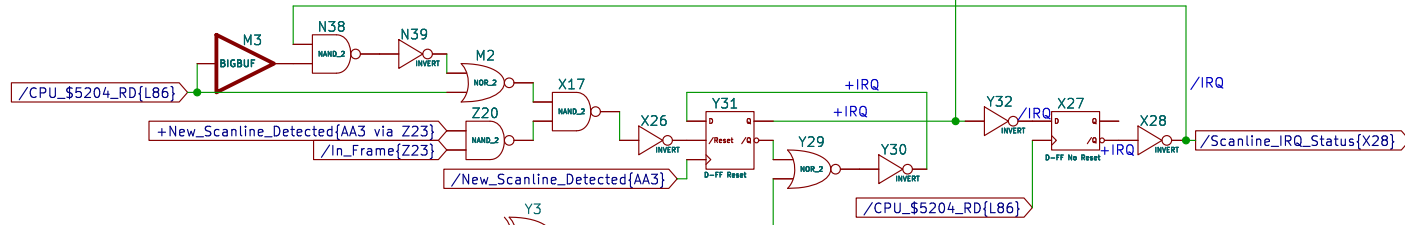
Notes:

- Note: The differential clockbursts' outputs are flipped in these tribufs, which makes it have +DE instead of /DE.
- Note: When the value written is \$00, this loops back the existing value instead of accepting the value \$00.

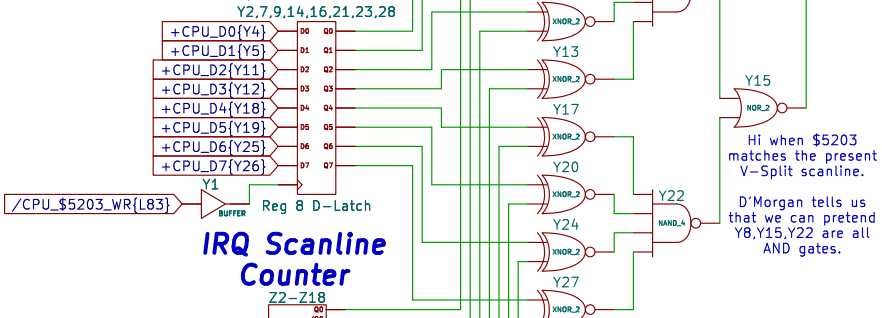
Title:		
Size: USLedge	Date:	Rev:
KiCad E.D.A. 9.0.9		Id: 9/25



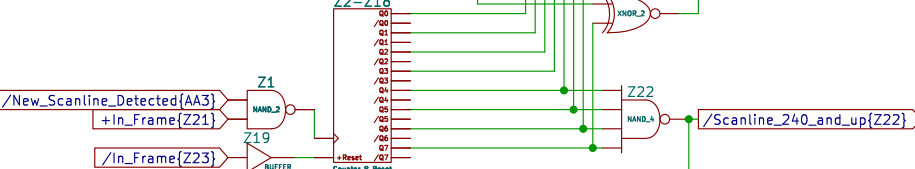
\$5204 Register Scanline IRQ Enable



\$5203 Register IRQ Scanline Compare Value



IRQ Scanline Counter

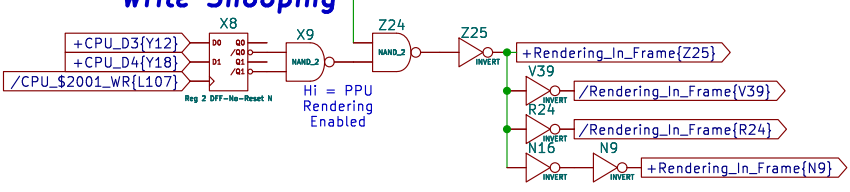


\$2002 Register Read Snooping

Latches \$2002.7 value read by CPU for 1 CPU cycle.

\$2001 Register Write Snooping

Hi = PPU Rendering Enabled



Sheet: /Scanline Detection/
File: scanline_detection.kicad_sch

Title:

Size: B Date: Rev: KiCad E.D.A. 9.0.9 Id: 10/25

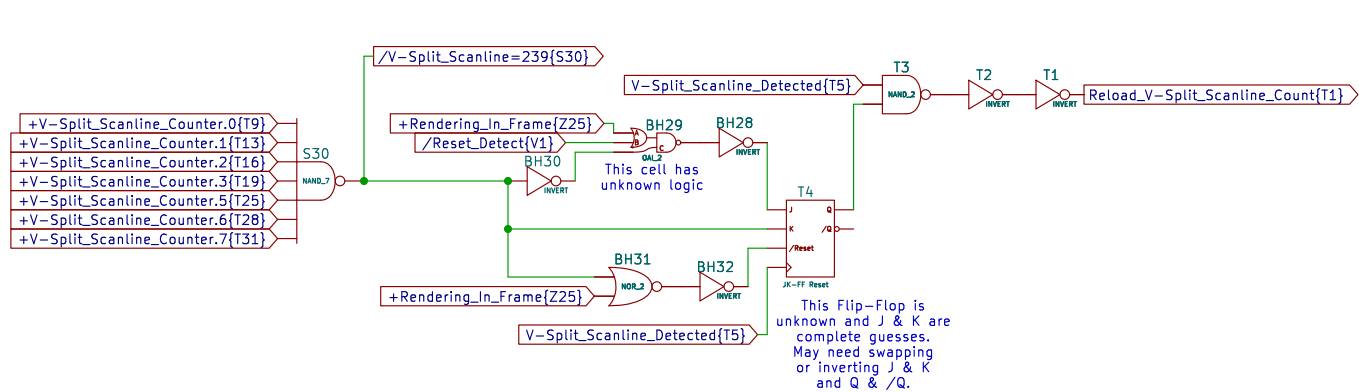
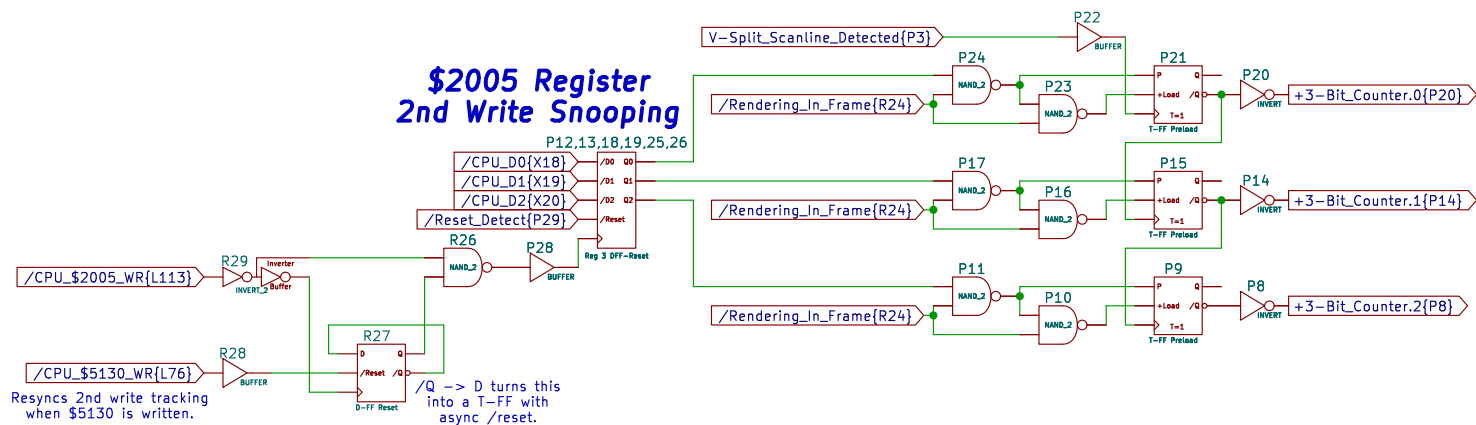
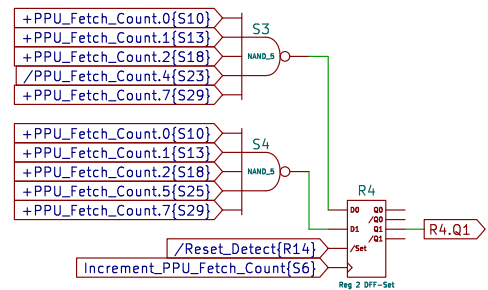
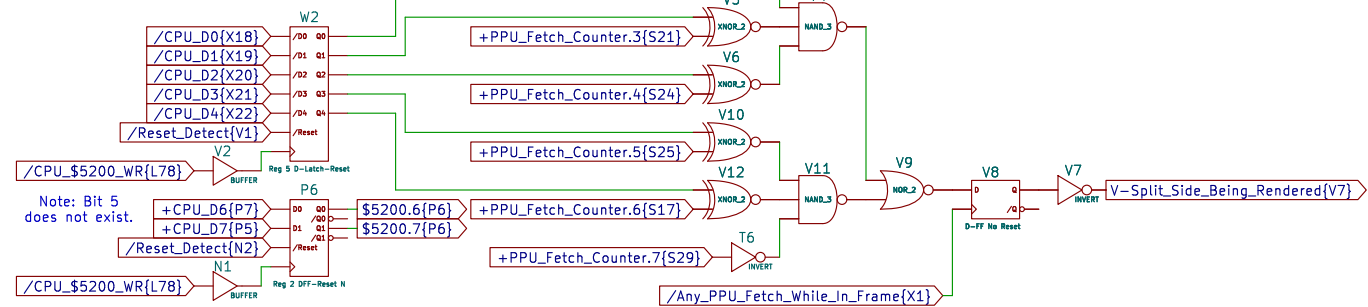
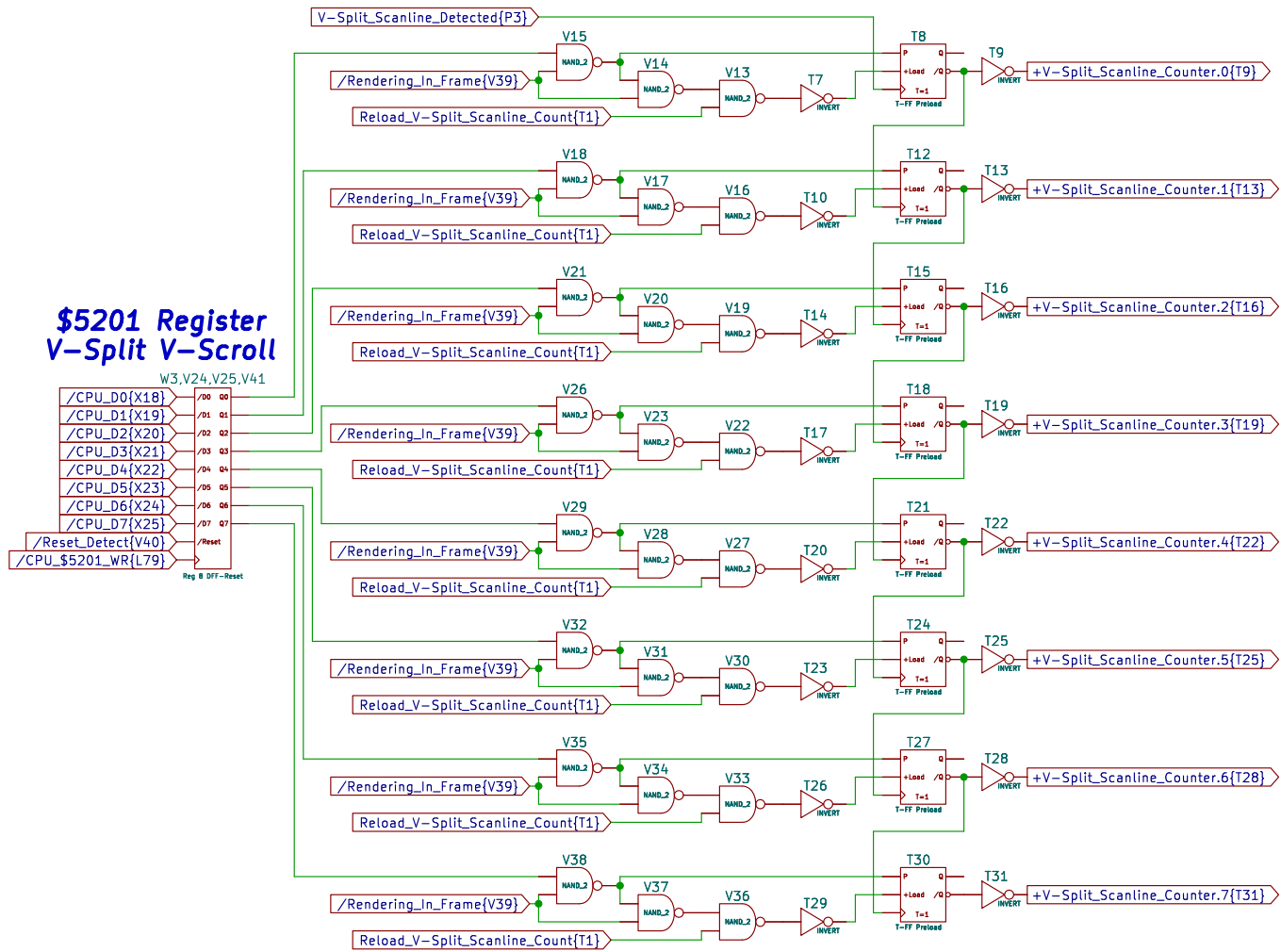


Diagram illustrating the PPU Fetch Counter circuit, showing the internal logic and output signals.

PPU Fetch Counter

The circuit consists of several components and signals:

- Inputs:**
 - `Increment_PPU_Fetch_Count{S6}`
 - `/New_Scanline_Detected{R2}`
 - `V-Split_Scanline_Detected_Internal{R8}`
 - `/Reset_Detect{R14}`
- Logic and Counters:**
 - Counter 0:** `S8` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.0{S10}`.
 - Counter 1:** `S12` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.1{S12}` and `+PPU_Fetch_Counter.1{S13}`.
 - Counter 2:** `S15` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.2{S18}` and `/PPU_Fetch_Counter.2{S15}`.
 - Counter 3:** `S20` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.3{S21}`.
 - Counter 4:** `S23` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.4{S24}` and `/PPU_Fetch_Counter.4{S23}`.
 - Counter 5:** `S26` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.5{S25}` and `/PPU_Fetch_Counter.5{S26}`.
 - Counter 6:** `S27` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.6{S17}` and `/PPU_Fetch_Counter.6{S27}`.
 - Counter 7:** `S28` (JK-FF Preset) with `+Load` and `/Q` outputs. `+Load` is connected to `/New_Scanline_Detected{R2}` and `/Reset_Detect{R14}`. `/Q` is connected to `+PPU_Fetch_Counter.7{S29}` and `/PPU_Fetch_Counter.7{S28}`.
- Control Logic:**
 - `R9` (NAND.2) and `R10` (NAND.2) are connected to `V-Split_Scanline_Detected_Internal{R8}` and `/Reset_Detect{R14}`.
 - `R11` (NAND.3) and `R12` (NAND.2) are connected to `V-Split_Scanline_Detected_Internal{R8}` and `/Reset_Detect{R14}`.
 - `R13` (NAND.3) is connected to `V-Split_Scanline_Detected_Internal{R8}` and `/Reset_Detect{R14}`.
 - `S22` (NAND.3) is connected to `V-Split_Scanline_Detected_Internal{R8}` and `/Reset_Detect{R14}`.

Output Signals:

- `+PPU_Fetch_Counter.0{S10}`
- `+PPU_Fetch_Counter.1{S12}`
- `+PPU_Fetch_Counter.1{S13}`
- `+PPU_Fetch_Counter.2{S18}`
- `/PPU_Fetch_Counter.2{S15}`
- `+PPU_Fetch_Counter.3{S21}`
- `+PPU_Fetch_Counter.4{S24}`
- `/PPU_Fetch_Counter.4{S23}`
- `+PPU_Fetch_Counter.5{S25}`
- `/PPU_Fetch_Counter.5{S26}`
- `+PPU_Fetch_Counter.6{S17}`
- `/PPU_Fetch_Counter.6{S27}`
- `+PPU_Fetch_Counter.7{S29}`
- `/PPU_Fetch_Counter.7{S28}`

Additional Notes:

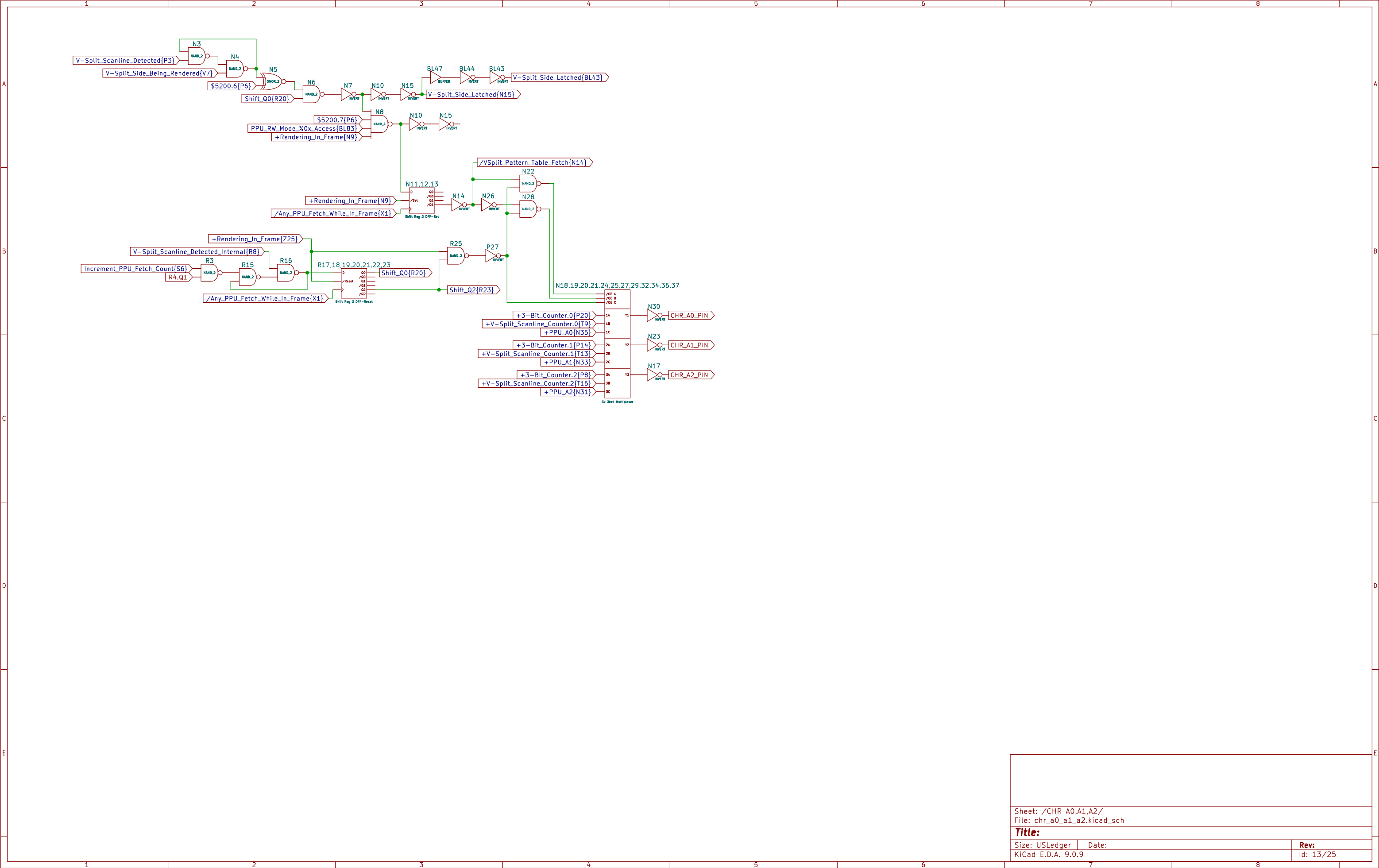
- `S16` (NOR.8) is connected to `+PPU_Fetch_Counter.0{S10}`, `+PPU_Fetch_Counter.1{S12}`, `/PPU_Fetch_Counter.2{S15}`, `+PPU_Fetch_Counter.3{S21}`, `/PPU_Fetch_Counter.4{S23}`, `/PPU_Fetch_Counter.5{S26}`, `/PPU_Fetch_Counter.6{S27}`, and `/PPU_Fetch_Counter.7{S28}`. Its output is `Value 244 or 11`.
- `R5` (JK-FF Reset) is connected to `/New_Scanline_Detected{R2}`, `/Reset_Detect{R14}`, and `Increment_PPU_Fetch_Count{S6}`. Its output is `R5.Q`.

Annotations:

- I think this may be a NAND 8 instead of NOR.
- This Flip-Flop is unknown and J & K are complete guesses. May need swapping or inverting J & K and Q & /Q.

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\$5104 Register ExRAM Mode

The schematic illustrates the internal logic of the \$5104 Register ExRAM Mode. It shows the connection of CPU and PPU data buses to the ExRAM mode, including control signals like /Reset_Detect and various logic gates (NAND, OR, XOR, AND, NOT) and buffers. The diagram includes a title block at the bottom right with the title 'Title:', size 'USLedger', date, and revision 'Rev: 14/25'.

**Sheet: /ExRAM Mode/
File: exram_mode.kicad_sch**

Title:

Size: USLedger Date:

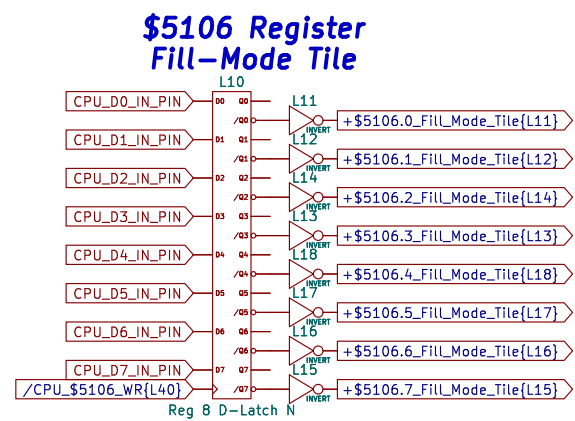
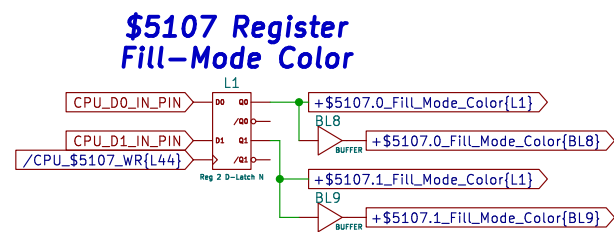
KiCad E.D.A. 9.0.9

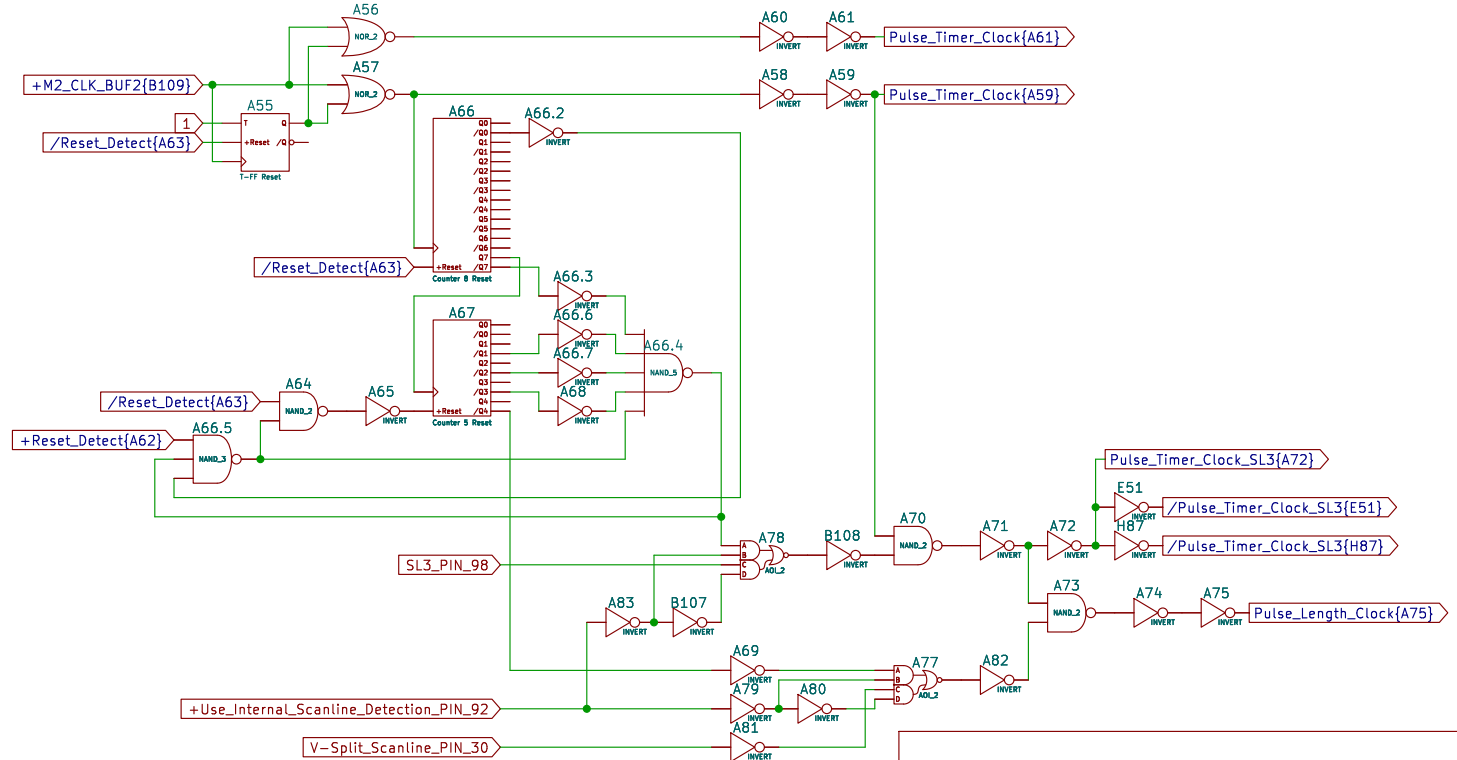
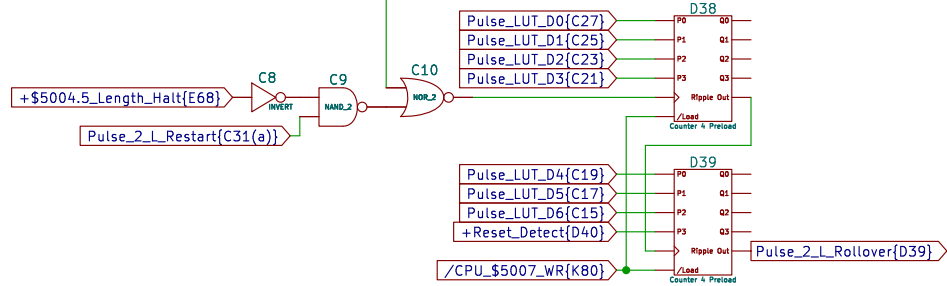
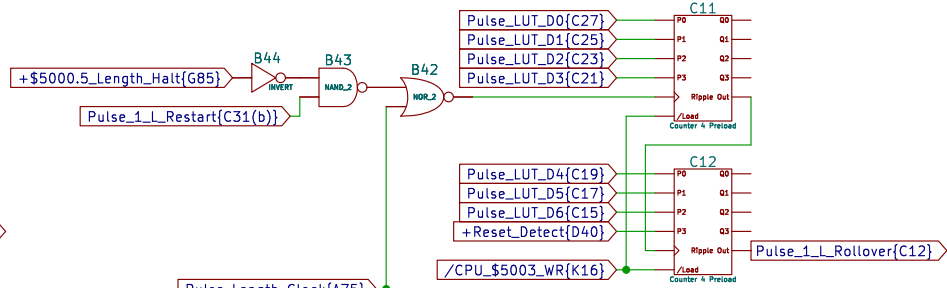
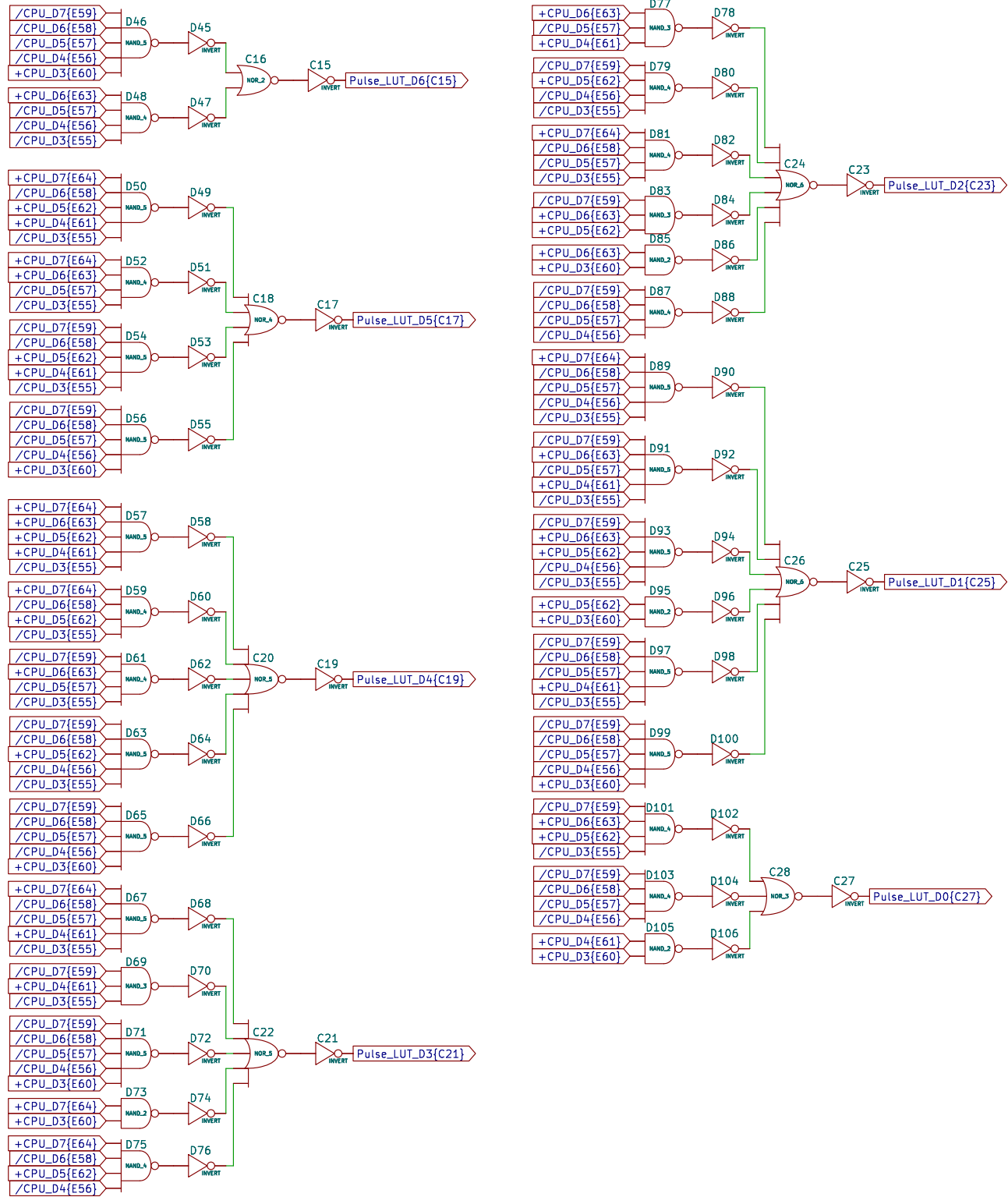
Rev: 14/25

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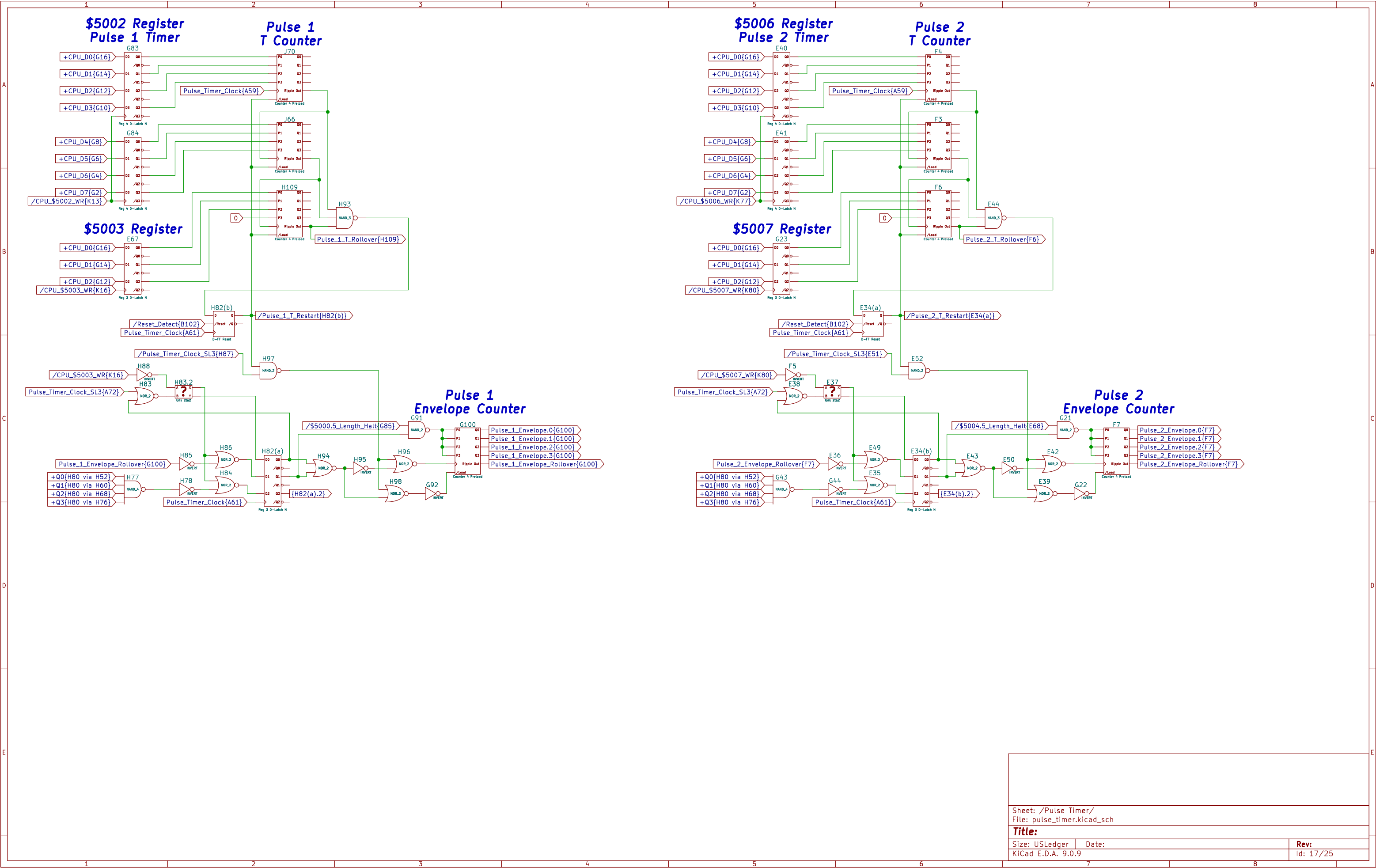
Rev:

d: 14/25

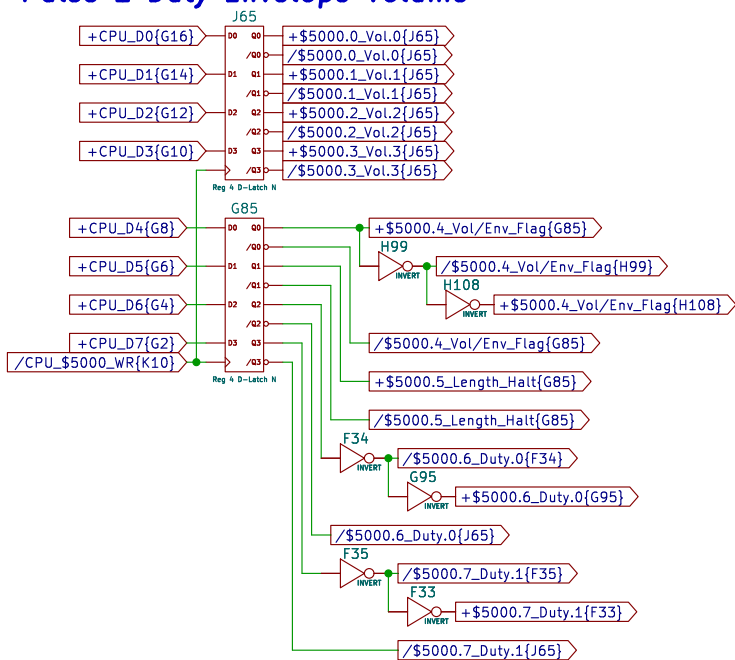




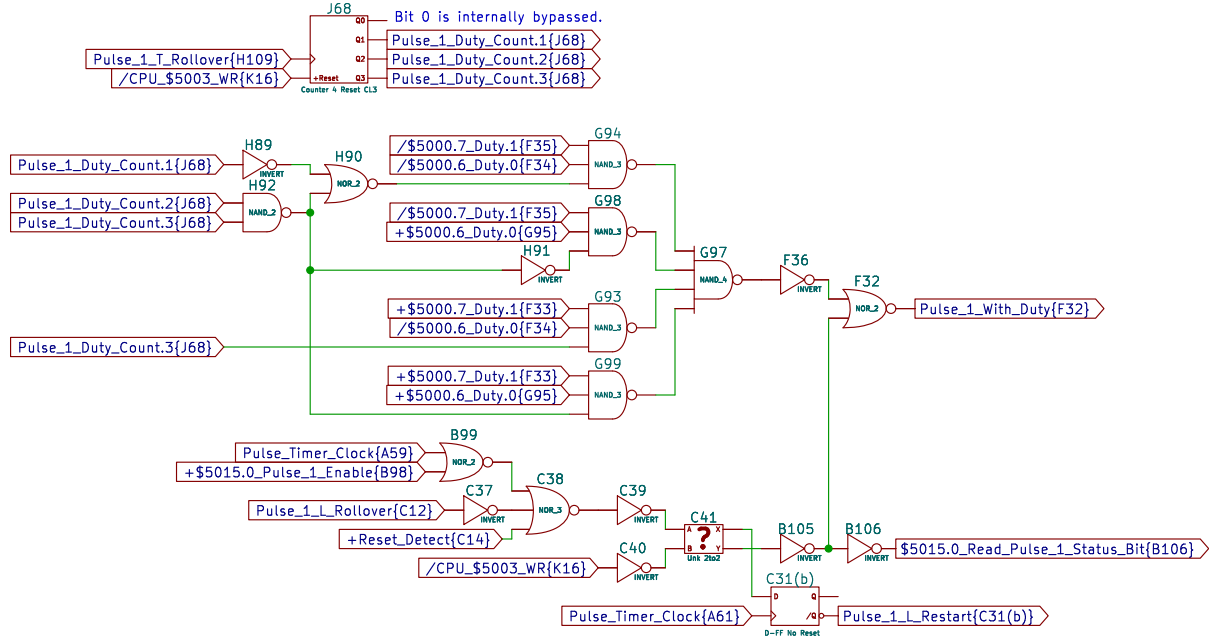
Size: USLedger	Date:	Rev:
KiCad E.D.A. 9.0.9		Id: 16/25



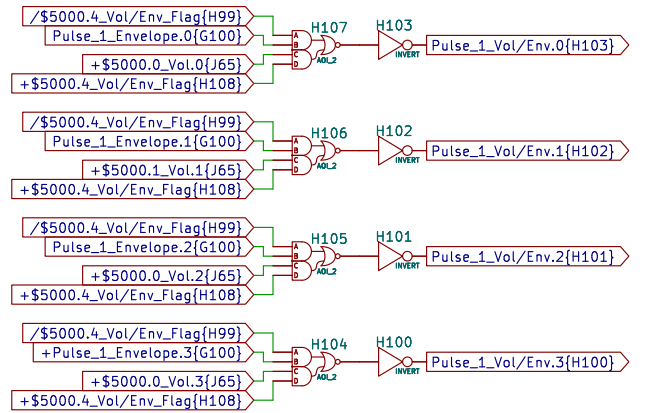
\$5000 Register Pulse 1 Duty Envelope Volume



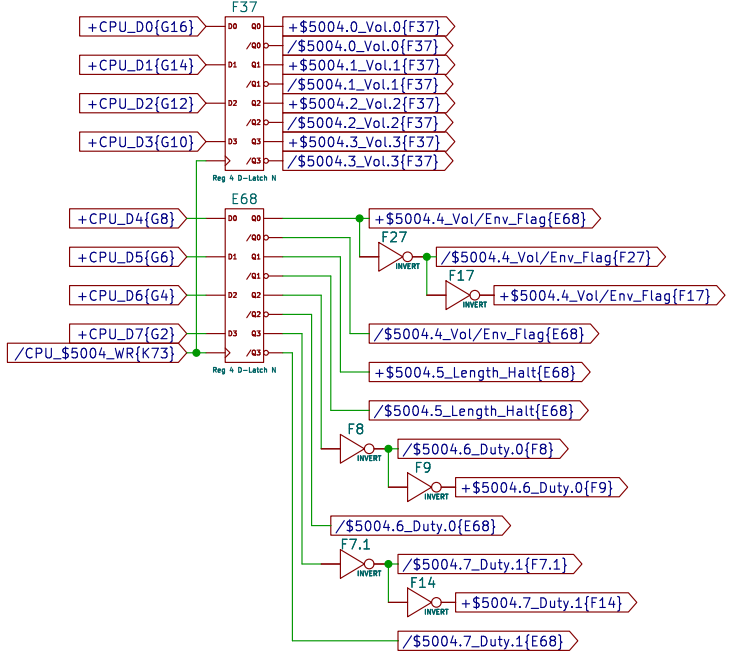
Pulse 1 Duty Cycle Counter



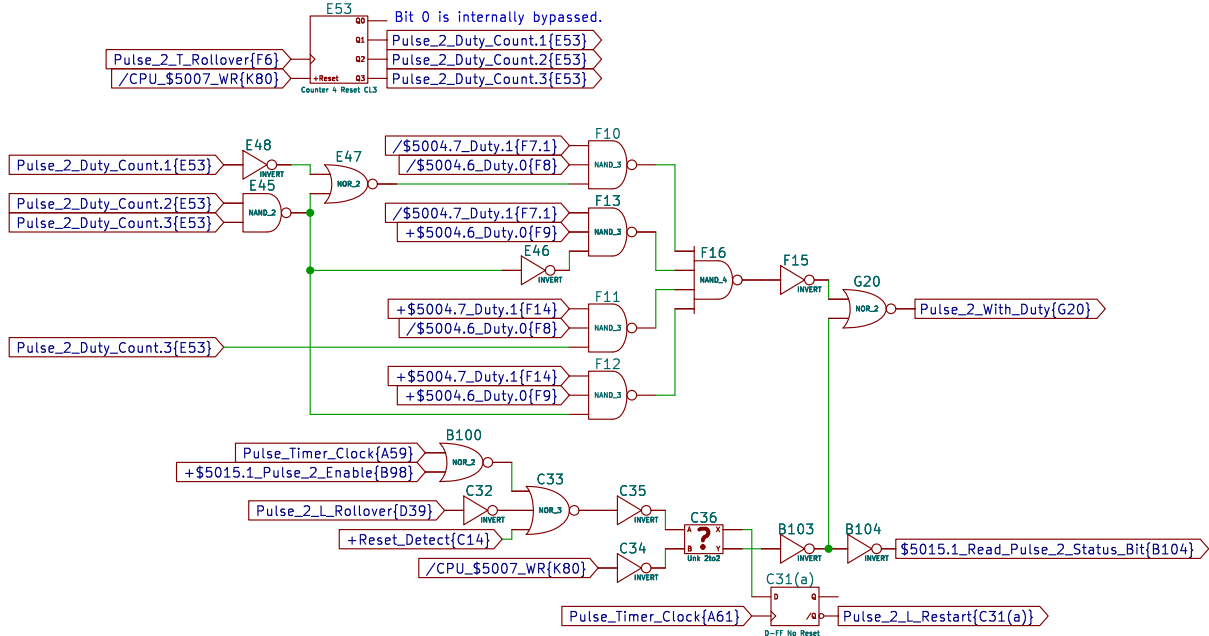
Pulse 1 Envelope/Volume Multiplexing



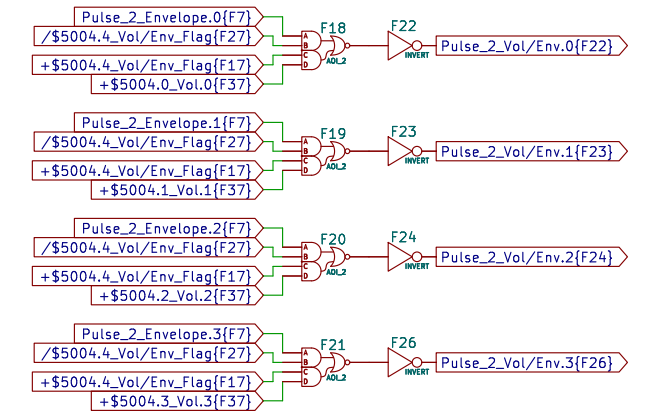
\$5004 Register Pulse 2 Duty Envelope Volume



Pulse 2 Duty Cycle Counter



Pulse 2 Envelope/Volume Multiplexing



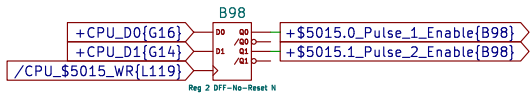
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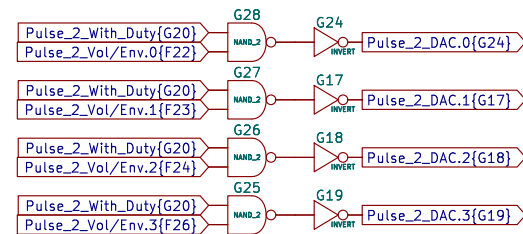
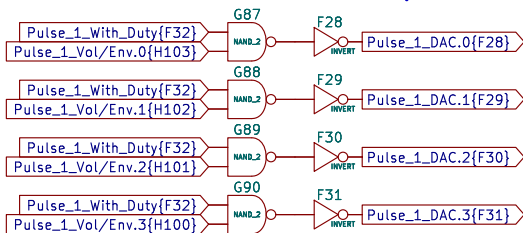
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KiCad E.D.A. 9.0.9

Rev:
Id: 18/25

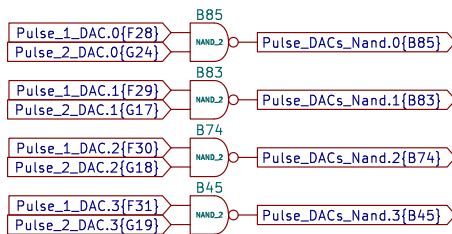
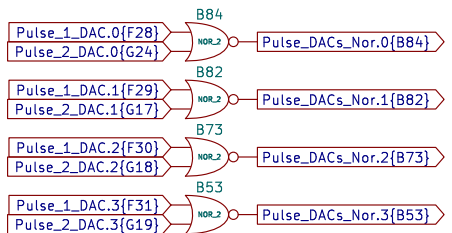
\$5015 Write Register Pulse 1&2 Status



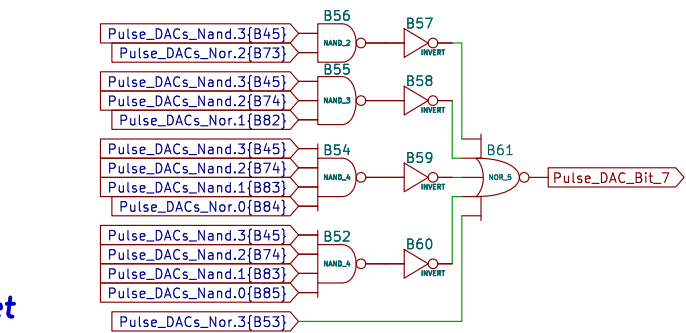
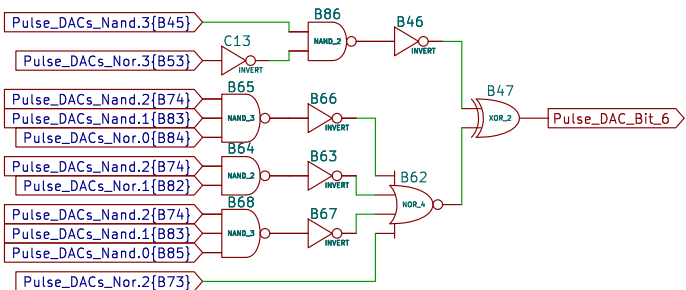
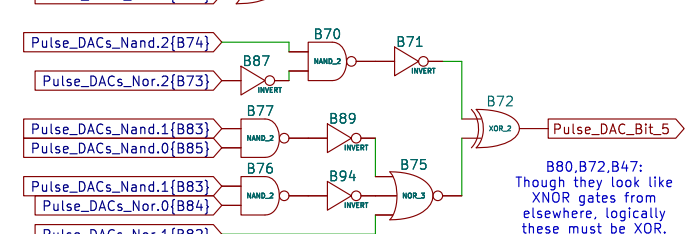
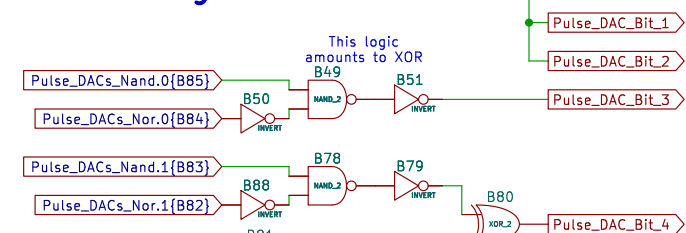
Pulse 1&2 Individual 4-bit DAC Outputs



Components for 4-bit Adder



4-bit Adder Adding Pulse 1&2 DACs



Don't know what this is yet
But it is important.

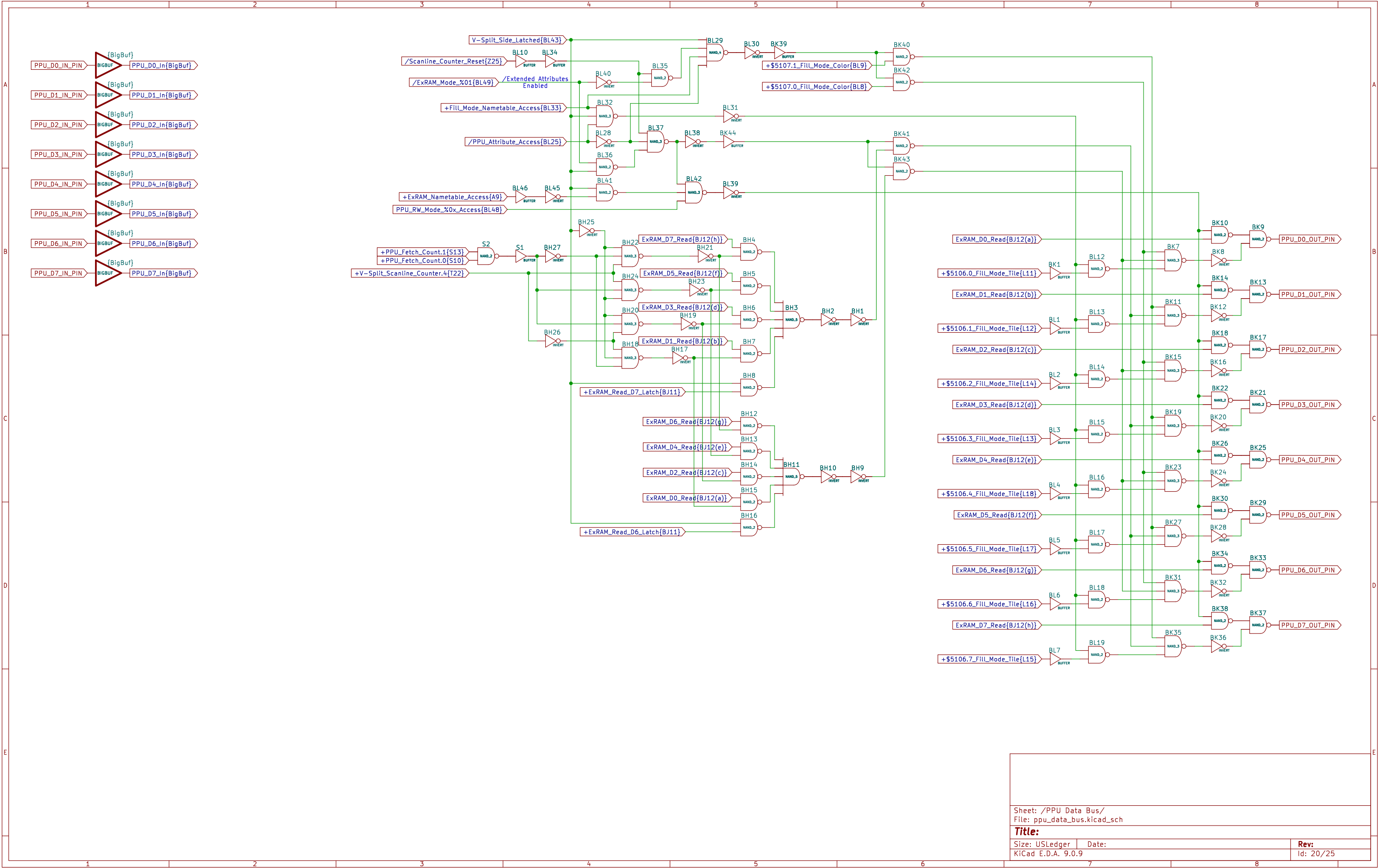
Don't know what this is yet
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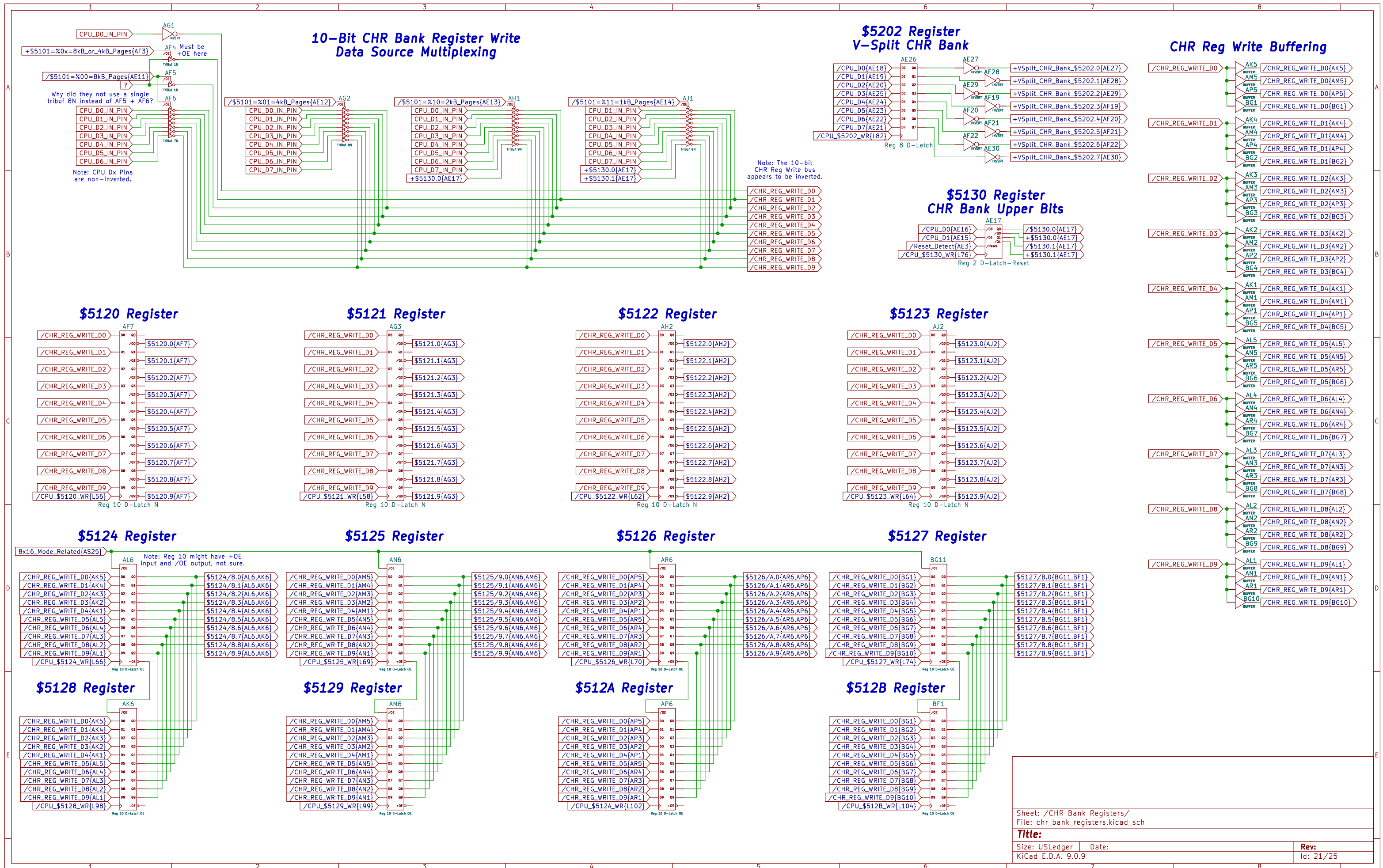
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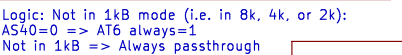
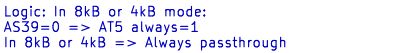
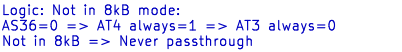
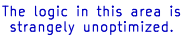
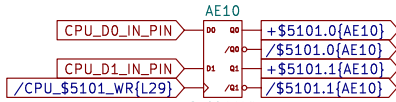
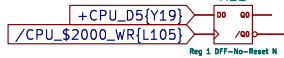
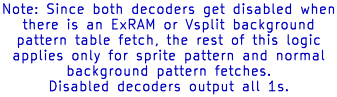
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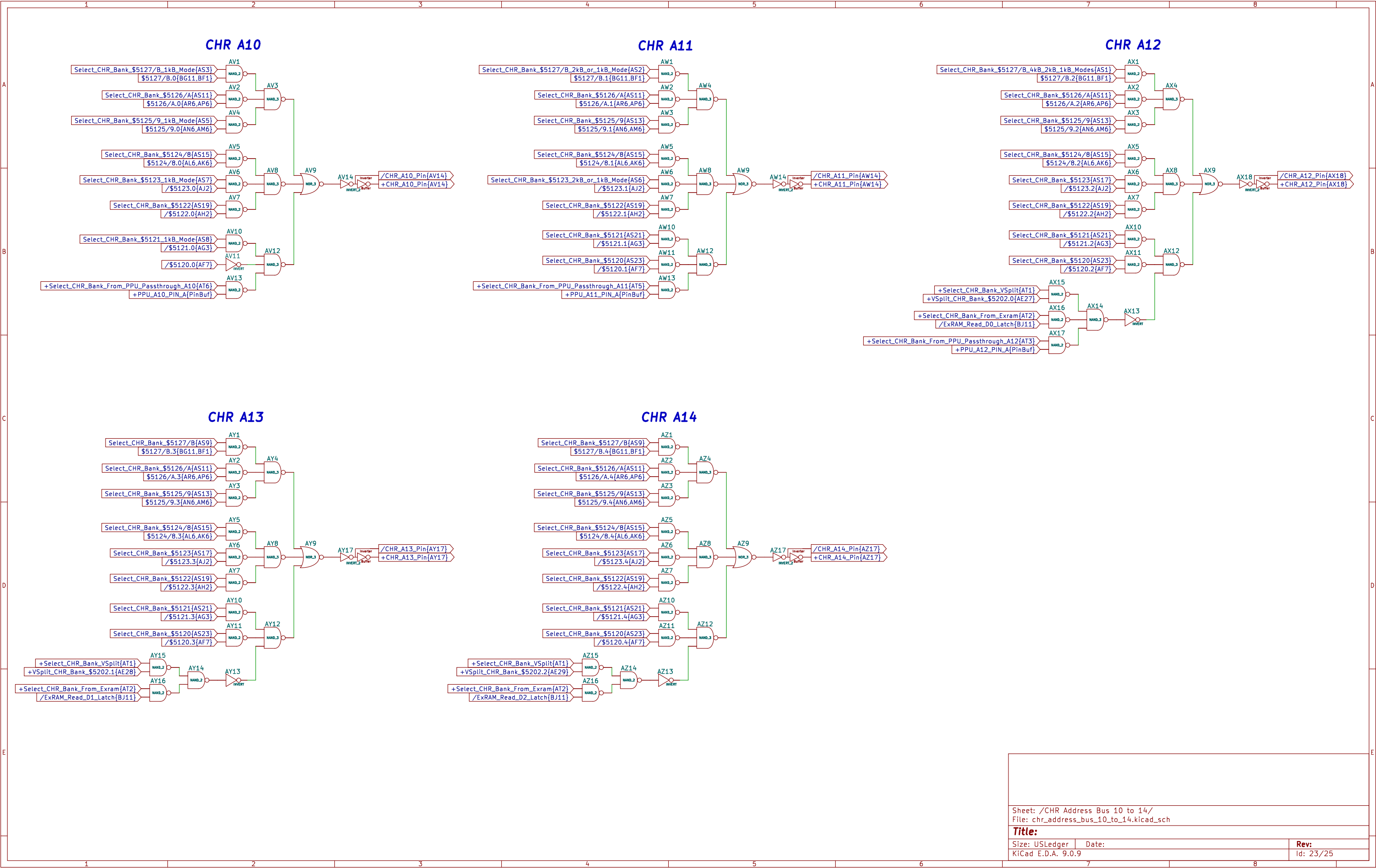
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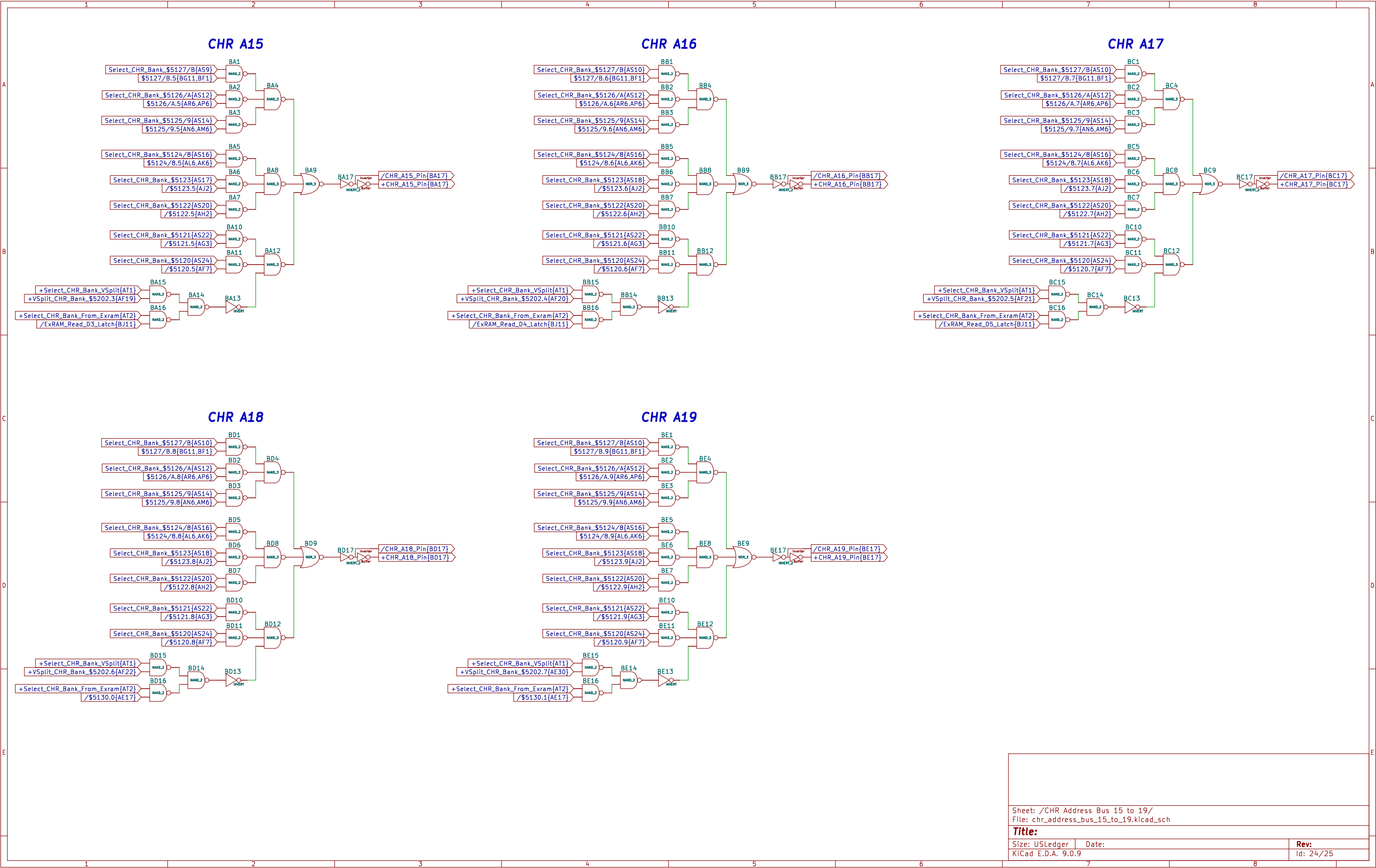
Rev:
Id: 19/25











This sheet contains all residual items that don't serve any apparent purpose.

